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Assignment 1: AI System Specification

ElderCare AI – A Mobile App for Elderly Emotional and Health Support

Chosen System:

An AI-powered mobile application designed to help elderly people who feel lonely or need daily healthcare assistance. The app features an emotional chatbot, medicine reminders, mood detection, SOS alerts, and activity suggestions.

Task 1: PEAS Specification

Performance Measures:

How well users take their medicines on time (adherence rate)

Improvement in user's mood over time

How much users interact with the app and their satisfaction level

How fast and successful the emergency SOS feature works

Environment:

The app works on elderly people's smartphones in their homes. The environment includes changing moods, different health conditions, family members, and daily routines.

Actuators:

Voice and text replies from the chatbot

Push notifications for reminders

SOS emergency calling and messaging system

Suggestions for yoga or light physical activities

Sensors:

Microphone for voice input and emotion detection

Emoji and text input for mood checking

Phone accelerometer for activity tracking

Camera (optional for face expression with permission)

GPS and usage history

Task 2: Environment Classification

Partially Observable: The app cannot fully understand the user's real emotions or health condition all the time. It only gets information from voice, emojis, or what the user tells it.

Multi-agent: The system works with the elderly user, their family members, and emergency services.

Stochastic: Human emotions and health situations are unpredictable.

Sequential: Every conversation or reminder affects the user's future mood and health.

Dynamic: The user's mood and needs keep changing throughout the day.

Continuous: Emotions, voice tone, and activity levels are continuous, not in fixed steps.

Task 3: Critical Analysis

Which dimension is the biggest challenge?

Partially Observable is the most difficult part. It is hard to correctly understand an elderly person's emotions because they may not speak clearly or may not want to share everything. The AI has to work with limited and unclear information.

Utility Function:

$$U = 0.45 \times \text{Emotional_Wellbeing} + 0.25 \times \text{Safety_Adherence} + 0.2 \times \text{Routine_Management} - 0.1 \times \text{User_Fatigue}$$

If we double the weight of Emotional_Wellbeing, the app will focus more on friendly conversations and emotional support rather than always pushing reminders. It will act more like a caring friend.

Task 4: Structured Representation (JSON)

JSON{

 "system_name": "ElderCare AI - Companion App for Elderly",

 "peas": {

 "performance_measures": ["Medication adherence rate", "Mood improvement score", "User engagement & satisfaction", "Emergency response success rate"],

```

"environment": "Elderly users' homes through smartphones with changing moods and health conditions",
"actuators": ["Voice & text chatbot", "Push notifications", "SOS alerts", "Activity suggestions"],
"sensors": ["Microphone", "Emoji input", "Accelerometer", "Camera (optional)", "GPS & history"]
},
"environment_classification": {
  "observability": {"choice": "Partially Observable", "justification": "Cannot fully see user's real emotions and health"},
  "agents": {"choice": "Multi-agent", "justification": "Works with user, family and emergency services"},
  "deterministic": {"choice": "Stochastic", "justification": "Emotions and health are unpredictable"},
  "episodic": {"choice": "Sequential", "justification": "Interactions affect future user wellbeing"},
  "static": {"choice": "Dynamic", "justification": "User mood and needs change all the time"},
  "discrete": {"choice": "Continuous", "justification": "Emotions and activity levels vary continuously"}
},
"utility_function": "U = 0.45*Emotional_Wellbeing + 0.25*Safety + 0.2*Routine - 0.1*Fatigue"
}

```